

# Chapter 4

## Lectures 10.1 and 10.2

### Lagrange's Theorem

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In this lecture, we will introduce some new object called cosets, which are needed to prove Lagrange's Theorem. We will see some examples and study their properties.

## The GOAL is to prove Lagrange's Thm:

**THEOREM 4.4. LAGRANGE'S THEOREM.** *Let  $H$  be a subgroup of a finite group  $G$ . Then the order of  $G$  is a multiple of the order of  $H$ .*

## What are cosets?

**Definition 4.1.** Let  $H$  be a subgroup of a group  $G$  and  $a \in G$ . The set  $Ha = \{ha \mid h \in H\}$  is called the **right coset** of  $H$  in  $G$  **generated** by  $a$ , while  $aH = \{ah \mid h \in H\}$  is called the **left coset** of  $H$  in  $G$  **generated** by  $a$ .

### Remarks

Observe that in **additive notation** the left and right cosets are:

$$H \pm a = \{h \pm a \mid h \in H\}, \quad a \pm H = \{a \pm h \mid h \in H\}$$

It is important to notice that **cosets in a group  $G$  is a subset of  $G$ .**

**Cosets are not elements in  $G$ !**

**Example 4.2.** Find all the left and right cosets of  $H = \{\varepsilon, (1\ 3)\}$  in  $S_3$ .

Recall that  $S_3 = \{\varepsilon, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$ . Then we have

Take  $a = (1\ 2)$ .

$$aH = (1\ 2)H = \{(1\ 2), (1\ 2)(1\ 3)\} \stackrel{\oplus}{=} \{(1\ 2), (1\ 3\ 2)\}$$

$$\stackrel{\oplus}{(1\ 2)(1\ 3)} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} = (1\ 3\ 2)$$

$$Ha = H(1\ 2) = \{(1\ 2), (1\ 3)(1\ 2)\} = \{(1\ 2), (1\ 2\ 3)\}$$

$$(1\ 3)(1\ 2) = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = (1\ 2\ 3)$$

Exercises. Compute  $H(2\ 3)$  and

$$(2\ 3)H.$$

**Example 4.3.** Find all the cosets of the subgroup  $4\mathbb{Z}$  in the additive group  $\mathbb{Z}$ . In this case, since  $\mathbb{Z}$  is an abelian group, it is enough to compute the right (or the left) cosets.

For  $a \in \mathbb{Z}$ , we want to compute the left (or right) generated by  $a$ :

$$4\mathbb{Z} + a = a + 4\mathbb{Z}$$

$n := 4q + a$ ,  $q \in \mathbb{Z}$   $\longrightarrow$   $a$  is the remainder when  $n$  is divided by 4.

Notice that  $a$  is the remainder when  $n$  is divided by 4. Recall that there are only 4 possible remainders when divided by 4: 0, 1, 2, 3.

Thus, we only have 4 cosets:

$\overline{0} = 4\mathbb{Z} + 0 = \{ \dots, -12, -8, -4, 0, 4, 8, 12, \dots \}$

$\overline{1} = 4\mathbb{Z} + 1 = \{ \dots, -11, -7, -3, 1, 5, 9, 13, \dots \}$

$\overline{2} = 4\mathbb{Z} + 2 = \dots$

$\overline{3} = 4\mathbb{Z} + 3 = \dots$

$$-3 = 4(-1) + 1$$

## Some important properties of cosets

Let  $a \in G$  and  $H$  a subgroup of  $G$ . Then:

1.  $H1 = H = 1H$ , that is,  $H$  is a right and left coset of itself.
2. It is obvious that  $a \in Ha$  and  $a \in aH$  since  $a = 1 \cdot a = a \cdot 1$  and  $1 \in H$ .
3. If  $G$  is abelian, then  $Ha = aH$  for all  $a \in G$  and all subgroups  $H$  of  $G$ .  
For any  $h \in H$ ,  $ha = ah$
4. 3 might not be true if  $G$  is not abelian.

Take  $G = S_3$  and  $H = \{e, (1\ 3)\}$ ,  $a = (1\ 2)$

$$(1\ 2)H \neq H(1\ 2).$$

5. If  $a \in Hb$ , then  $Ha = Hb$ . ( $b \in G$ )

Since  $Ha$  and  $Hb$  are sets, we need to show that

$$Ha \subseteq Hb \text{ and } Hb \subseteq Ha.$$

From  $a \in Hb$ , we get that  $a = hb$  for some element  $h \in H$ . Take  $x \in Ha$ , say  $x = h_1 a$ , for some  $h_1 \in H$ , and let us show that  $x \in Hb$ .

$$\underline{x} = h_1 a = h_1 (h b) = \underbrace{(h_1 h)}_e b \in \underline{Hb} \checkmark$$

To prove that  $\underline{Hb} \stackrel{?}{=} \underline{Ha}$ , use that  
 $\boxed{b = h^{-1}a}$

→ **6.**  $\underline{Ha = Hb}$  if and only if  $ab^{-1} \in H$ .

**7.**  $Ha = H$  if and only if  $a \in H$ . (Take  $b = 1$ )

In additive notation **6** and **7** become

$$H + a = H + b \Leftrightarrow a - b \in H; \quad H + a = H \Leftrightarrow a \in H$$

For instance,  $6\mathbb{Z} + 7 = 6\mathbb{Z} + 1$  since  $7 - 1 = 6 \in 6\mathbb{Z}$ .

Proof of ⑥

$\Leftarrow$  | Assume that  $ab^{-1} \in H$ . Then  $ab^{-1} = h$  for some  $h \in H$ . From here we obtain that  $a = hb \in Hb$ . Applying ⑤ we get that  $\underline{Ha = Hb}$ .

$\Rightarrow$  | Suppose now that  $\underline{Ha = Hb}$ . Then  $a = 1 \cdot a$ ,  $1 \in H$   
 $\underline{a} \in \underline{Ha} = \underline{Hb}$ , and we can find  $h \in H$   
 such that  $\underline{a = hb}$ . But then  $\underline{ab^{-1} = h} \in H$ ,  
 as required.

8. There exists a bijective map from  $H$  to  $Ha$ . In particular, if  $H$  is finite, then any coset  $Ha$  has the same number of elements as  $H$  and therefore all the cosets have the same number of elements.

Define  $\varphi: H \longrightarrow Ha$   
$$h \longmapsto \varphi(h) = ha$$

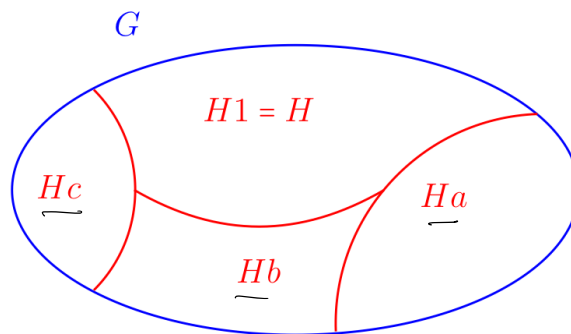
This map is surjective because any element of  $Ha$  is of the form  $\underline{h_1}a$ ,  $h_1 \in H$ , and  $\varphi(h_1) = h_1a$ .

$\varphi$  is injective:  $\varphi(h) = \varphi(h') \stackrel{?}{\Rightarrow} h = h'$

$\varphi(h) = \varphi(h') \Rightarrow \underline{ha = h'a} \Rightarrow \underline{h = h'}$  ✓

↑  
Cancellation  
law in  $G$   
or multiplying  
by  $\underline{a^{-1}}$

9. The family of all the cosets of  $H$ , as  $a$  ranges over  $G$ , is a partition of  $G$ . It means that  $G = \cup_{a \in G} Ha$ , and  $Ha \neq Hb$  implies  $\underline{Ha \cap Hb = \emptyset}$ .



Notice that  $G$  is the union of all the cosets since for any  $g \in G$ , we have  $g \in Hg$  (because  $g = 1 \cdot g$ ).

Take  $Ha$  and  $Hb$  two cosets of  $H$  in  $G$ . If  $Ha \cap Hb = \emptyset$ , then we are done. Otherwise, let  $Ha \cap Hb \neq \emptyset$  and take  $x \in \underline{Ha} \cap \underline{Hb}$ . Then  $x = \underline{h_1 a} = \underline{h_2 b}$  for some  $h_1, h_2 \in H$ .

From here we obtain that  $\swarrow$  Property ⑤

$$\underline{a = (h_1^{-1} h_2) b} \in \underline{Hb} \Rightarrow \underline{Ha = Hb} \quad \text{😊}$$



**THEOREM 4.4. LAGRANGE'S THEOREM.** Let  $H$  be a subgroup of a finite group  $G$ . Then the order of  $G$  is a multiple of the order of  $H$ .

Proof Suppose that  $Ha_1, Ha_2, \dots, Ha_k$  are all the different cosets of  $H$  in  $G$ . Then

$$G = Ha_1 \cup Ha_2 \cup \dots \cup Ha_k.$$

On the other hand since it is a partition, the union is disjoint and so

$$\underline{|G|} = \underline{|Ha_1|} + \underline{|Ha_2|} + \dots + \underline{|Ha_k|} \stackrel{\text{Property 8}}{=} \underline{|H|} + \underline{|H|} + \dots + \underline{|H|}$$

$$= |H| + |H| + \dots^k + |H| = \underline{k |H|}$$

This shows that the order of  $G$  is a multiple of the order of  $H$ , as required.



**COROLLARY 4.5.** The order of any element of a finite group divides the order of the group.

Proof. Let  $G$  be a finite group and take  $g \in G$ . By definition the order of  $g$  is the order of the cyclic subgroup  $\langle g \rangle$  generated by  $g$ . By Lagrange's Theorem we know that  $|\langle g \rangle| \mid |G|$ , that is  $o(g) \mid |G|$ . 

**COROLLARY 4.6.** Every group of prime order is cyclic.

Proof. Let  $p$  be a prime and suppose that  $G$  is a group of order  $p$ . Take any  $g \in G$  such that  $g \neq 1$ . Consider the cyclic subgroup generated by  $g$ :  $\langle g \rangle$ . Notice that  $\langle g \rangle \neq \{1\}$  because  $g \in \langle g \rangle$  and  $g \neq 1$ .

By Lagrange's Theorem we obtain that

$$|\langle g \rangle| \mid |G| = p.$$

From here we obtain that  $|\langle g \rangle| = p$ .  
 This implies  $G = \underline{\langle g \rangle}$  and  $G$  is  
cyclic. ☺

Definition. Let  $G$  be a group and  $H$   
 a subgroup of  $G$ . The INDEX of  $H$  in  
 $G$  is the number of cosets of  $H$  in  $G$ .  
 We denote it by  $|G:H|$

Example.  $|\mathbb{Z}:4\mathbb{Z}| = 4$

• If  $G$  is finite, then looking at the proof of Lagrange's then we obtain that

$$|G:H| = \frac{|G|}{|H|}.$$

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